

Coexistence From First Principles

The kernel and reconstructed structure of Madhyasth Darshan

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The thesis. This paper gives a layered formal reconstruction of Madhyasth Darshan’s ontological core. Its minimal kernel contains three primitives — the omnipresent, actionless medium *satta*, real countable units, and their saturation in that medium — together with two organizing commitments. Saturation provisions every unit with activeness, mutual recognisability, and inherent regulation; every unit is also oriented toward development and completeness appropriate to its order. An order-indexed complementarity relation then describes how units couple, fulfil, and sometimes close into a new bounded whole. Units are definite bearers, but their four inseparable aspects are not modelled as four pre-filled scalar properties: form is real and reflected in mutual facing; *guna* is power or effect in mutuality; *svabhava* is the order- and context-specific essentiality of that effect; and *dharma* is inseparable innateness evidenced in fulfilment. This kernel makes explicit a structure the primary texts state across many contexts without claiming that the texts themselves derive everything from these axioms.

The reconstruction shows how that structure changes rather than merely repeats across the four orders. Compound-closure is literal in the physicochemical line, compositional in the biological line, and an acknowledged conduct analogy in humane social organization. Time is represented as the duration of unit-activity, natural and excited states admit a conditional stability model, and constitutional completeness remains a guarded transition rather than a universal attractor. At the knowledge order the model adds the reflexive activity of *jeevan*: knowing, believing, recognising, fulfilling, evaluating, and transmitting must cohere. Karma makes understanding answerable to result and conduct; faculty harmonies give non-scalar form to happiness, peace, contentment, and bliss; and the five dimensions of universal order connect individual awakening to social evidence.

The paper’s main achievement is therefore not a proof that Madhyasth Darshan is true, nor a reduction of all its teachings to one formula. It is a checkable map of what depends on what. The recovery audit records each notion’s textual provenance, role within the model, and epistemic standing; the mathematical spine clarifies composition, duration, stability, and social compatibility while marking where analogy ends. The result is a compact architecture that unifies the darshan’s ontological, sentient, and social layers while leaving its taught data, se-

lection predicates, metaphysical commitments, and empirical questions visible rather than hiding them inside the formalism.

1. One schema, across the orders

Madhyasth Darshan is usually read through its human programme — justice in relationships, family-based order, humane education. The texts support a stronger reading: the general structure is asserted first, and the human case is organized as its most demanding instance. *Samadhanatmak Bhautikvad* opens in fully general terms —

“Each unit is a whole along with its environment.”

— SB p. 13–14

“Each unit is orderliness with its ness and participates in overall orderliness.”

— SB p. 13–14 (*ness = self-ness in the bilingual edition; PDF extraction drops the prefix*)

— and *Manav Vyavahar Darshan* states the same universality: “All units saturated in the Omnipresence (permeative and transparent) have form, properties, essential nature & dharma, and have inherent orderliness & participate in overall orderliness” (MVD p. 11). *Manav Karm Darshan*, whose title-page principle is simply “Effort – Motion – Result” (KD, Principle), develops the same ground as a science of activity from the atom upward.

So the task is not extrapolation from humans to atoms but **articulation**: stating, in one structure, what the texts assert piecewise about atoms (MVD p. 8; JV p. 67), cells (JV p. 82), organisms (SB p. 16), and human collectives (MVD p. 55, p. 161). The whole of it can be held in one picture before any formalism:

- **Ground.** There is one omnipresent, actionless medium, *satta*, and there are countless real units. Every unit is *saturated* in the medium — soaked, submerged, surrounded. That saturation, and nothing added to it, makes each unit energy-replete and law-legible, so each unit is **active**: its activity is effort, motion, and result together.
- **Couple.** Units couple by **complementarity** — one’s result entering the other as input, never competition. Complementarity takes an order-specific form; its material instance is a unit short of completeness meeting one in surplus (deficiency and surplus).
- **Close and recur.** When a complementary material coupling closes a new boundary, it is a new definite bearer, one tier up. Its constitution and order are new, and its four aspects become evident through the new unit’s mutualities. The closure schema reappears in typed forms: literal *yaugik* compound-closure in the physicochemical line, compositional organization in the biological line, and an explicit conduct analogy for individuals organizing into families and undivided society.

- **Run reflexively.** At the fixed point where a composite atom reaches constitutional completeness, its activity can take *activity itself* as its object. That reflexive turn is *jeevan*, the sentient unit. All five faculties and ten activities belong to *jeevan*. An animal body limits their expression; a developed human body supports their full expression, while delusion (*bhram*) can misdirect evaluation by identifying the self with the body.

One closure schema organizes the tiered structure; at constitutional completeness a distinct **reflexive** move (§2.3) opens the knowledge order. A worked instance runs through the whole paper in miniature: two atoms, one hungry and one overfull (MVD p. 8), couple by complementarity, close into a molecule with a new conduct (MVD p. 42), which re-enters the same schema in order-specific form toward the planetary and biological lines; carried up the development progression to constitutional completeness, the same activity turns reflexive and is *jeevan* (SB p. 61).

Precision earns its keep by letting the structure be **carried to systems Shri Nagraj did not himself analyse**. Forward, as design: an organisation, an economy, or an institution is **humane and internally self-sustaining** exactly when it satisfies each reconstructed clause — identifiable units, definite couplings with explicit expectation profiles, capacity sufficient for value to flow, composition by complementary need, persistence conditional on fulfilment, and an explicit carrier of transmission across member turnover. (This is a criterion for humane, coercion-free order — not a law of empirical survival.) In reverse, as diagnosis: a failing system is walked clause by clause until the unmet one is found, and the knowledge-order discontinuity predicts that human-tier systems fail first at the understanding-dependent steps.

Three cautions govern what follows. The texts *assert* piecewise; this paper *reconstructs* — every reconstruction is this paper's construction, warranted by citations but not identical with them. The formalism is a **lens for clarity**, not a proof machine: it proves nothing about the reality of the medium or of *jeevan*, which are exactly where the darshan parts from mainstream science. And a reduction can flatten: the kernel keeps units co-eternal with the medium — real carriers, not clusters distilled from activity — and refuses to reconstruct away what the texts present as taught content.

Model status. Every claim carries a status along **two independent axes**, and the paper is explicit about both. **Textual provenance** — is the claim a **textual assertion** (stated in MVD/SB/JV/KD), a **formal reconstruction** (the paper's own kernel-level organisation — κ , the coupling anatomy), or a **mathematical analogy** (the categorical spine of §6, the physical readings)? **Epistemic standing** — is it **established within the model** (a result of the reconstruction), a **taught datum** (content the texts fix and the model cannot produce), or an **open metaphysical posit** (asserted, not established)? The two axes cross: *satta* and *jeevan* are textual assertions *and* open metaphysical posits; κ is a formal reconstruction established within the model; the six value-kinds are textual assertions that remain taught data. The audit's role-tags (*primitive*, *axiom*, *definition*, *theorem*, *data*) locate a notion *within* the reconstruction — so a “theorem” there is a result *within the model*, not a

proof about the world. The kernel **organizes** the darshan's structure; it does not claim the texts derive it.

2. The kernel

2.1 Three primitives

O (*satta*, Omnipresence) — one, real, all-pervasive, **actionless**, state-complete, and energy- and knowledge-natured (*gyan*): “Space is uniform in all dimensions... this reality is omnipresent” (JV p. 31); state-complete while everything else is state-dynamic (MVD p. 12). O is not a unit, and no statement quantifying over units applies to O.

U (units, *ikai*) — real, co-eternal, countable **carriers**: “Nature is in the form of entities that can be counted” (JV p. 45). A unit bears a boundary, constitution, order, state, capacities, and activity; it is never built from activity, occasions, observation, or process. These bearer-conditions make mutual effects definite without turning the unit into an isolated four-property record. Carrier-primacy is a commitment, not a convenience: a unit is not identified either with its behaviour or with the relations through which that behaviour becomes evident (§6, §10).

Saturation — the one ground relation: every unit is soaked, submerged, and surrounded in the single O, with “no provision in existence to separate units of nature from Omnipresence” (JV p. 18). Saturation is constitutive co-location in a shared medium, the *first* ontological relation, distinct from any unit-to-unit coupling, and not a physical drawing from O (SB pp. 49, 62).

Activeness is deliberately **not** a primitive. The texts state its status as an identity, “Unit + Energy fullness = Activeness” (SB p. 69) — a co-presence conferred by saturation, recovered below as a theorem (§2.4), not a fourth building block.

2.2 Two axioms

A1 (Endowment — the *satta* firewall). Through saturation, O endows each unit's carrier with three distinguishable but inseparable invariants. **(a) Activeness:** inexhaustible energy manifests as the triad of effort (*shram*), motion (*gati*), and result (*parinam*) — “There is no unit or particle in nature which is not replete with energy... This activeness is seen in every unit in the form of effort, motion and result” (SB p. 69; KD 3.1). **(b) Recognition-capability:** activity is unit-to-unit legible, since “there is provision in every unit for recognising one another based on law” (SB p. 57). **(c) Inherent regulation:** that lawfulness disciplines a unit's activity in its natural state and appears at successive orders as structural-, seed-, species-, and *sanskar*-conformance (§4.6). Recognition and regulation are two consequences of lawful intelligibility, not one undifferentiated capacity: the former opens mutuality, the latter maintains definite conduct. In the material, pranic, and animal orders, reflection, relational evaluation, recognition, and fulfilment are definite under the given constitution and conditions; fallible mis-evaluation is not projected backward into insentient nature. The fire-

wall is part of the axiom: **O is never an input to, nor a summand of, a unit's activity** — it provisions energy, legibility, and regulation without push, update term, or depletion. The nucleus within each atom, and mediative *atma* within *jeevan*, localise regulation without O ever becoming an agent (MVD p. 26).

A2 (Completeness-orientation). Every unit's activity is directed toward **development and completeness** — “each unit, embedded in the complete, is active and developing solely for completeness” (SB p. 51) — and not toward **indefinite quantitative increase**. This orientation is universal, but its **terminal triadic form** is tiered. Along the atomic development line it culminates at **constitutional completeness** — “every atom continues to undergo insertion and expulsion of particles until constitutional completeness” (SB p. 71), the threshold at which *jeevan* appears (T1). The fully-realized terminal goals — **restfulness of effort, destination of motion, immortality of result** (SB p. 58) — are the awakening targets of *jeevan* (activity- and conduct-completeness, T2–T3); the first three orders realize completeness as development toward order, not as consciously-attained restfulness. The darshan does not treat indefinite quantitative increase as the telos; this reconstruction therefore represents completeness as a **qualitative terminal condition**, not an unbounded scalar objective (§8).

2.3 The generative move

κ (compound-closure). Given saturated units whose capacities are **complementary** in a definite mutual context — in the material instance a deficiency met by a surplus, with complementarity taking an order-specific form at higher tiers (§7) — their activities couple and, when the coupling closes a *new* motion-path, present one bounded whole: a new unit, one tier up, whose components “discard their own conducts, and present another kind of conduct” (MVD p. 42). Written as a rule:

$u_1, \dots, u_n \in U_o$ saturated in O motion-path	$C_o(u_1, \dots, u_n)$	closure of a new
$\kappa(u_1, \dots, u_n) \in U$ — a new bearer and relational profile, +1 tier		

Here U_o is the type of units at order o , c collects the relevant state and environmental conditions, and $C_o(u_1, \dots, u_n; c)$ is the **order-indexed complementarity relation**. Deficiency met by surplus is the material instance; nutrient and seasonal fit, species and lineage fit, and understood value-expectation are higher-order instances (§7). The rule's premises are the **anatomy of a coupling**, spelled out in §4.3: **recognition** (ρ) identifies that C_o holds; **relational evaluation** (ϵ) fixes the definite fit and expectation profile under c ; **fulfilment** (ϕ) executes the coupling, with results flowing along that profile; and **closure** asks whether the executed coupling settles into one new bounded motion-path.

Thus $\kappa_o(u_1, \dots, u_n; c) = \text{closure}(\varphi_o(\rho_o(u_1, \dots, u_n; c), \varepsilon_o(u_1, \dots, u_n; c)))$. **Mixture** (*mishran*) is the same coupling when it does **not** close: φ still delivers results — “components all maintain their respective conducts” (MVD p. 42) — but no new boundary forms. Compound and mixture are the two closure outcomes of a fulfilled material coupling, which is why “composition is not development” (SB pp. 75–76). Because κ ’s output is again a unit, the closure schema reappears in an order-specific form — literal compound-closure in the physicochemical line, compositional organization in the biological line, and a declared formal analogy at the knowledge order (§4.4).

The reflexive turn. κ iterated along the development progression (*vikas-kram*) — “in the atom in the form of hungry and overfull atoms” (MVD p. 8) — reaches a distinguished fixed point: a composite atom whose result-aspect attains its A2 terminal, *constitutional completeness*, after which “both effort and motion become inexhaustible” (SB p. 61). At that fixed point the unit’s activity can take activity itself as its object. This reflexive re-application is **jeevan**. It supplies the faculty structure for conscious knowing and evaluation, but it does not make every *jeevan* fallible in the same way: animal conduct remains definite through a species-specific bodily medium; human mis-evaluation requires a human bodily medium together with *bhram*, the mistaken self-model in which *jeevan* identifies as the body (§4.8–§4.9). Reflexivity is a distinct move, not a second literal firing of κ ; it is **an interpretive model** of the knowledge order — this paper’s organising reading, not a derived theorem — flagged as such in the audit.

2.4 Corollaries

Activeness. By A1(a), every saturated unit is energy-replete and therefore active; its activity is the triad. “Unit + Energy fullness = Activeness” (SB p. 69) is thereby a theorem, and activeness is a *family indexed by units*, not a substrate they are built from. The triad’s internal constitution is a joint form — each moment the outcome of the other two (MVD p. 105) — and its combined activity is basic impulsion (*mool cheshta*, MVD p. 40). KD states each aspect’s sense directly: effort is the grandeur of the existent-state, motion the grandeur of spreading effect, result the grandeur of a different existent-state (KD 3.11).

Conservation. The texts assert existence is beginless and stable — “Existence neither increases nor decreases” (JV p. 127). Conservation of *activity* does not strictly follow from constancy of *existence*; the reconstruction takes it as a **model postulate**, consistent with A1’s inexhaustible source and A2’s non-annihilating orientation but warranted textually rather than derived.

Coupling and non-isolation. That units couple does not follow from shared saturation and activeness alone — independently active units need not interact. The texts supply the missing premise directly: “Nothing is isolated – that is the principle” (JV p. 43). The reconstruction therefore takes **no-isolation as a premise** (a textual assertion): every unit participates in at least one coupling, and universal coupling holds because of it — rather than separating coupling as a theorem from a residual totality assumption.

3. Notation: category theory in brief

The formal spine of §6 uses category theory; this section gives the working vocabulary in plain language so the paper is self-contained, and readers at home with it can skip ahead.

Category theory replaces “what is each thing made of?” with “how does each thing relate to everything else?”. An **object** is a thing to talk about; a **morphism** (arrow) is a relationship or a way from one object to another; **composition** chains arrows, and every object has an **identity** arrow. Two dual readings matter here. An **algebra** for a construction is a thing *built up* by a rule — the natural home of “generated”: the units of nature can be encoded as the algebra generated by κ , so definitions over units become definitions by recursion on how a unit was built. A **coalgebra** is the dual — a carrier together with a map that *unfolds* its behaviour without identifying the carrier with that behaviour; it keeps a unit a substance that bears an activity rather than a bundle of relations. A **colimit (gluing)** joins pieces into one whole along what they share — the shape of families joining into one society. These three — the generating algebra upward, the carrier-coalgebra downward, and the gluing colimit sideways — are all the machinery the spine needs; the richer constructions are collected as optional annotations (§6.3).

4. The reconstructed structure

The kernel organizes the notions below without claiming to generate all their content. Each notion is tagged **[def]** (a name for kernel structure), **[thm]** (a consequence within the reconstruction), or **[data]** (taught content the kernel cannot produce). The fuller provenance and epistemic audit appears in §5.

4.1 Definite bearers and relational manifestation

A **unit [def]** is a saturated carrier in U , active as the triad (§2.4), disciplined along the motion-path that is at once its boundary and line of regulation. Insentient units are “active within the bounds of their length, width and height”; sentient units are “active... beyond the bounds” (MVD p. 33) — the motion-path bounds the activity’s extent without bounding its reach. An **atom** (*parmanu*) is not the elementary particle of physics: its kind, state, and measure are fixed by the number of particles in its nucleus (*madhyansh*) and its orbiting particles (MVD p. 42; KD 3.1). Write its bearer profile as

$$B(u) = (\text{boundary, constitution, order, state, capacities})$$

$B(u)$ is definite before a particular observer describes it. It prevents the opposite error of turning the unit into an empty relational node. This definiteness does not mean that every effect and value is pre-populated as an isolated outcome: the bearer and its capacities are real, while the concrete reflection, effect, essentiality, and evidence are fixed only with the mutually facing unit and conditions specified. The four aspects name how this real bearer is

present and effective in coexistence; they are neither four detachable substances nor a static record exhausted by observer-independent scalar values.

For a mutual context $c = (u, v, \text{environment}, \text{time})$ admitted by c_o , define an **aspect-manifestation map [formal reconstruction, with data]**:

$$M(u, c) = (\text{refl}_c(\text{roop}_u), \text{eff}_c(\Gamma_u), \text{ess}_c(\text{eff}), \text{evid}_c(\text{dharma}_u))$$

Here roop_u is the unit's real bounded form, while $\text{refl}_c(\text{roop}_u)$ is its definite reflection or presentation in mutual facing. Γ_u is the unit's inseparable capacity profile; $\text{eff}_c(\Gamma_u)$ is *guna*, the power or effect manifested with a partner and conditions. $\text{ess}_c(\text{eff})$ is *svabhav*, the usefulness or essentiality of that effect according to state, motion, plane, order, and context. dharma_u is inseparable innateness and fulfilment; $\text{evid}_c(\text{dharma}_u)$ records its definite evidence in participation without reducing *dharma* to an observation. The map is indexed by mutuality, but its value is fixed by the bearers and conditions rather than created by the observer. The per-order expressions remain taught data:

Order	Order-specific expression of <i>svabhav</i>	<i>Dharma</i> (cumulative)
Material (<i>padarth</i>)	integration–disintegration	existence (<i>astitva</i>)
Bio (<i>pran</i>)	vitalising–devitalising	+ growth (<i>pushti</i>)
Animal (<i>jeev</i>)	cruel–uncruel	+ hope to live
Knowledge / human (<i>gyan</i>)	fortitude, courage, generosity, kindness, grace, compassion	+ happiness (<i>sukh</i>)

4.2 Relationships and value

Every unit-to-unit mutuality is a **coupling** — the general term, at every order. A **relationship** (*sambandh*) [def] is the knowledge-order coupling whose expectation profile $E(r)$ is definite “in the sense of completeness” between *jeevan*-bearing units (MVD pp. 61–62); it is not a preloaded property of either isolated party. An **association** is a coupling whose profile is voluntarily adopted. The lower orders have definite coupling and mutuality, not *sambandh* in this value-laden sense. The **value** of a unit in a coupling [thm-slot, with data] is what that coupling realises — its essentiality (*maulikta*), “the essentiality of units in every plane and order... It is values that are reciprocated and mutually recognised” (SB p. 50). That value is real at every tier because coupling is; *which* values fill the slot is taught data. The texts sort essentiality into six kinds — object values (utility and art) operating at all orders, and *jeevan*, human, established, and expression values operating only at the knowledge order and pre-supposing *jeevan* (MVD p. 306; JV pp. 43, 138–139). The nine established values flow from *jeevan* the moment a relationship is recognised — “when a mother has recognised her child

then motherly care starts surging by itself” (JV p. 138) — in a generative order (gratitude → affection, love, trust; JV p. 108), so V carries a partial ordering.

4.3 Recognition, fulfilment, and capacity

Recognition ρ [def] is A1(b) exercised between units — identification of a coupling and its profile. **Relational evaluation** ε [def] is the actual fit, effect, or congruence of the coupling under its definite bearers and conditions. **Fulfilment** ϕ [def] is the coupling executing: “Fulfilling is to evidence use, right-use and purposeful-use” (MVD p. 27, p. 62). In the material, pranic, and animal orders, ρ , ε , and ϕ are constitutionally definite; animal bodily limitation restricts expression without introducing mis-evaluation. At the human tier, **reflexive evaluation** μ is *jeevan*’s judgement of the relation and result; μ can agree with or misread ε (§4.9). This distinction lets the formalism affirm definite evaluation in *jada* without treating human normative judgement as a material process.

Fulfilment is modulated by **capacity [data]** — capacity (*kshamata*), ability (*yogyata*), and receptivity (*patrata*) (MVD p. 62): ϕ is realised at the level these permit, not by receptivity alone. In the material order the conversion of capacity and ability into effort is *ascending* (*agreshan*); its shortfall is frustration (*kshobh*), “incompleteness of receptivity” (MVD p. 79) — the triad failing to close toward restfulness, which at the human tier is “their yearning for restfulness” that drives awakening (MVD p. 104). Recognition, definite relational evaluation, and fulfilment are phases of every coupling; the generative move κ (§2.3, §4.4) is what a fulfilled coupling becomes when it closes. At the human knowledge order the additional comparison μ can complete the cycle $\rho \rightarrow \phi \rightarrow \mu \rightarrow \text{mutual satisfaction}$ as justice or distort it under delusion.

4.4 Composition: compound and mixture

Composition is the generative move κ (§2.3). **Compound** (*yaugik*) closes a new motion-path and presents a new definite bearer with a new constitution, order, and map of context-indexed manifestations; **mixture** (*mishran*) does not, and yields a non-generating aggregate whose components “all maintain their respective conducts” (MVD p. 42) — so mere enlargement never crosses an order plateau (“composition is not development”, SB pp. 75–76).

κ names literal *yaugik* compound-closure in the physicochemical line. The same formal schema organizes biological composition without asserting that every cell-to-organism organization is *yaugik* in exactly the atomic sense, and it is extended only analogically to human organization, where constitutionally complete humans organize into families, communities, and undivided society (MVD p. 55). The **sentience-creating** instance of closure — constitutional completeness (T1) — runs only along the atomic development progression (*vikas-kram*): new sentience is an atomic threshold (SB p. 92; KD 3.2–3.3), so composing *jeevan* yields an assembly of *jeevan*, never a larger *jeevan*. At the knowledge order composition is achieved: voluntary, understanding-mediated, and fallible, requiring shared cause, goal, programme, and transmission. Awakening progression (*jagriti-kram*) is not κ ; it runs within an already constitutionally complete *jeevan* toward T2–T3 (§4.8).

At the knowledge order the closure test acquires definite content. A human assembly is treated in the reconstruction **analogically as compound-like** — a **formal conduct-compound** — exactly when its collective coupling closes: shared cause and goal recognised (ρ), values fulfilled (ϕ), mutual satisfaction reached (μ — justice at scale, §4.9), and the method carried across member turnover by education-*sanskar* (τ , §4.5). Such an assembly **organizes as a whole that exhibits the four human goals — resolution, prosperity, fearlessness, and coexistence** — and persists as a self-continuing tradition in coexistence (MVD p. 55; SB pp. 246–247) — the *goal* of human assembly, its assembly-scale analogue of the awakening transitions T2–T3. An assembly held together by fear, accumulation, or enforcement fails the *humane* test — it is not self-reproducing without coercion and lacks internal orderliness — even where it survives empirically for a time by enforcement; the criterion (P3) concerns humane, coercion-free order, not a law of institutional survival. Two cautions keep the analogy honest: it is a **conduct-compound, not a sentience-compound** — no collective *atma*, since sentience-creating closure (T1) is atomic-only; and the texts support an *organized* family, community, and undivided society, not clearly a new *yaugik* unit in the same sense as a molecule — so “compound” here is a **formal analogy**, not a claim that the assembly is a fresh unit of nature. This is the exact sense in which composing *jeevan* yields an *organized assembly* of *jeevan*, never a larger *jeevan*. The chain is instantiated in [How To Form Self-Sustaining Organizations](#) and, at the scale of humankind, [How Undivided Society Is Established](#).

4.5 State, persistence, and transmission

A unit is in its **natural state [def]** when its couplings are fulfilled within conducive conditions and in an **excited state** otherwise: “Each unit moves towards development in its natural state and decline in its excited state” (SB p. 14). In triad terms the excited state is the activity failing to close toward its A2 target. **Persistence [thm, of humane order]** follows: an assembly’s *self-sustaining, coercion-free* order holds exactly while its couplings stay fulfilled and lapses when they do not — natural-state closure sustained. This is a criterion for humane, internally-reproduced order, not a necessary-and-sufficient law of empirical survival: a coercive assembly can endure for a time by enforcement. Conducive conditions are order-relative; in the material order even an excited state can be complementary — an expelled particle feeds a hungry atom (SB p. 15), a fulfilled coupling (ϕ) that delivers results without closing into a new unit — while in the deluded knowledge order excitation harms both self and others (SB p. 16).

Transmission τ [def, with data] re-instantiates an assembly’s method of composition (*rachna vidhi*) in new member-units, “inherent in the pran sutra situated in the cell” (JV p. 82). Its form — a result whose content is a method, recurring across member turnover — is generated; its carriers are taught data, ascending with the orders (constitution / seed / lineage / education-*sanskar*; JV p. 48; MVD p. 93). At the knowledge order τ carries *evidenced* understanding only: “ignorance cannot flow in tradition” (JV p. 49).

4.6 Orders and the regulation ladder

U is partitioned into four **orders [def, with data]** — material, pranic/bio, animal, knowledge (JV p. 47; MVD p. 9) — each a persistence regime differing in how a unit’s method survives:

Order	Expression through the order	Regulation regime	Mode
Material	reflected form, effect, relational evaluation, recognition, fulfilment	result-/structural-conformance	definite
Pranic / bio	+ selection through living composition	seed-conformance (<i>beej</i>)	definite
Animal	<i>jeevan</i> expressed through a species-limited body	species/lineage-conformance (<i>vansh</i>)	definite; bodily expression limited
Knowledge	all ten <i>jeevan</i> activities expressible through a developed human body	<i>sanskar</i> -conformance	fallible under <i>bhram</i> ; definite in awakening

Grounding: “The material order is regulated by structural conformity... the biological order by seed conformity... the animal order by species conformity... Humans achieve definite conduct through *sanskar*-conformity” (JV p. 48; SB p. 236). The **regulation ladder [def]** is the reading order of A1 and the orders, not new axioms: saturation provisions regulation *in* the unit; regulation becomes evident as law (universal across orders); the same law appears as order-specific conformance (definite in the first three, achieved in the knowledge order); within constitutionally complete *jeevan*, mediative *atma* disciplines the faculties (§4.8); and justice is the knowledge-order closure (§4.9). Orderliness at the order level is self-regulation (*swatah-saspurt*), inherent in co-existing orders, not dispensed by O acting as governor.

4.7 Planes and completeness transitions

Alongside the orders, SB names four **planes (*pad*)** — physicochemical, delusional, deific, and divine — as stages in nature’s progression toward completeness. The three completeness stages **[thm-orientation, with data]** are the fulfilled moments of the triad and map to plane transitions:

Transition	Completeness	From → to
T1	Constitutional (<i>gathanpurnata</i>)	Physicochemical → delusional
T2	Activity (<i>kriyapurnata</i>)	Delusional → deific
T3	Conduct (<i>vyavaharpurnata</i>)	Deific → divine

T1 is the reflexive fixed point of §2.3 — irreversible at the atomic level (SB p. 92); T2 and T3 are awakening milestones within constitutionally complete *jeevan*. Four progressions must not be collapsed: existential (*niyati-kram*, the fixed order emergence material → pranic → animal → knowledge), way of existence (*niyati-vidhi*, definiteness of each order's conduct), development (*vikas-kram*, through the physicochemical complex until T1), and awakening (*jagriti-kram*, within *jeevan* toward T2–T3).

Vikas-kram and *jagriti-kram* are one engine in **two regimes**, divided by the reflexive turn. Below it, a unit's completeness is **constitutional** and its deficiency is met from outside — surplus particles supplied by complementary units through κ . At and above it, constitution and the full faculty inventory are already complete, so the remaining deficiency is **realizational**: the faculties are not coherently exercised in understanding and conduct. This is met not by acquiring or creating a missing faculty but by *jeevan*'s activity turning inward and resolving *bhram*. The drive is the same completeness-orientation (A2) carried by the same triad — *kshobh*, the “frustration of effort [that] itself is their yearning for restfulness” (MVD p. 104), is what A2 feels like once the activity has become reflexive. What changes across the turn is only the currency: from particles-and-bonding to understanding-and-alignment. So “the constitution is already complete, what develops?” has a clean answer — realisation and coherent exercise develop, not constitution or faculty inventory.

4.8 Jeevan and the reflexive turn

Jeevan [def, with data] is the unit at the reflexive fixed point (§2.3): a constitutionally complete composite atom (*gathanpurna parmanu*) whose result-aspect has reached immortality, after which “both effort and motion become inexhaustible” (SB p. 61). Its constitution is the atom's nucleus-and-orbit structure — “The nucleus of the sentient unit... is referred to as *atma*. The particles in its first orbit... *buddhi*, the second... *chitta*, the third... *vritti*, the fourth... *mun*” (MVD p. 78) — operating through the projection–reflection cycle (*paravartan–pratyavartan*; KD 3.12): selection and taste in *mun*, analysis and deliberation in *vritti*, visualisation and contemplation in *chitta*, resolve and enlightenment in *buddhi*, authenticity and realisation in *atma* (JV p. 92; MVD p. 13). All five faculties and ten activities belong to every *jeevan*; their inventory is taught data, while reflexivity is the kernel's structural claim. Since *atma* is mediative, the faculties and body are naturally regulated by it, and effort resisting that regulation is discord (MVD pp. 77, 277) — inward regulation parallel to the atomic nucleus.

A human and an animal are joint forms of body and *jeevan*. The body is an expression medium, not the producer or owner of the faculties. Formalise the interface as a partial expression map

$$\text{express} : \text{Jeevan} \times \text{Body} \times \text{Context} \rightarrow \text{BodilyResult}$$

whose available outputs are constrained by the body's sensory, neural, and motor organization. The map is asymmetric: *jeevan* acts and knows through bodily states, while the body

conditions what can be evidenced. A species-specific animal body exposes a restricted subset of the ten activities; a normally developed human body supports their full exercise. Neurological immaturity, injury, or impairment may further restrict expression without changing the faculty inventory of *jeevan*. Which *jeevan* animates which body, and when, remains a selection predicate the kernel hosts but does not fix — taught data, at the same tier as the T1 predicate (§10).

What differs between an animal and a human is therefore not which faculties *jeevan* possesses but which activities the bodily medium can express and organise. Animal conduct remains species-conformant and definite because the animal medium does not support the full reflexive human circuit. The normally developed human medium does. The specifically human failure is *bhram*: *jeevan* takes the body to be the self and organises its existing resolve, desire, thought, and hope around bodily comfort, accumulation, and safety. The familiar “four and a half activities” description therefore names effective organisation under delusion, not the absence or dormancy of the remaining activities. Formally, bodily limitation is a restriction on `express`; delusion is an error in the self-model and evaluative routing. They must not be represented by the same gate.

Realizational development (§4.7) is the increasing coherence of activities already present. Each activity, **if it is justified** — if the reflexive cycle closes as justice, $\rho \rightarrow \phi \rightarrow \mu \rightarrow$ mutual satisfaction (§4.9) — evidences the *jeevan* values, happiness and peace; misjudged activity does not. Study supplies the proposal and understanding; practice tests, evidences, and stabilises that understanding in work, behaviour, and participation. Practice does not manufacture faculties or awaken them as previously absent mechanisms. Death and continuity are treated in the planned *Death, Continuity, and Rebirth* study; here the dyad enters only as the asymmetry that reflexive judgement and its conduct-evidence belong to *jeevan*, never to the body alone (JV p. 39).

4.9 Evaluation, justice, and the human goals

Reflexive evaluation μ [thm] is the human *jeevan*’s comparison of value delivered against the definite relational evaluation ϵ and value inherent in a relationship. The actual mutuality is not made indefinite by this second-order judgement. The model locates error as

$$\text{MisEval}(j, b, r, c) \quad \text{iff} \quad \text{HumanBody}(b) \wedge \text{Bhram}(j, b) \wedge \mu_j(r, c) \neq \epsilon(r, c)$$

The body provides the medium, *bhram* supplies the mistaken self-model, and μ is the judgement that can diverge. Without the *HumanBody* $\wedge$ *Bhram* conditions, sentience alone does not entail fallibility: animal recognition and fulfilment remain species-definite. The texts’ human scope remains: “Jeevan evaluates the values that emanate from itself... No external instrument is needed” (JV p. 139); “Values and evaluation are processes of understanding; they are not bodily or mechanical” (JV p. 39). Over-, under-, and mis-evaluation (MVD p. 38) are thus failure modes of deluded human reflexive judgement, not of material mutuality or of *jeevan* considered apart from its bodily medium.

Justice (*nyaya*) [**def**] is the closed reflexive cycle — “Recognising relationships, fulfilling values, evaluating, and achieving mutual satisfaction is justice” (MVD p. 311; JV p. 55): the core cycle $\rho \rightarrow \phi \rightarrow \mu \rightarrow$ mutual satisfaction, an operator over value, not a member of it. Its truth condition includes $\mu = \varepsilon$: the human judgement agrees with the definite mutual value actually fulfilled. This is the knowledge-order coupling’s full anatomy — recognition, fulfilment, evaluation — so compound-closure κ at the human tier (an organisation, §4.4) fires exactly when this cycle closes collectively. **Trust** (*vishwas*) [**data**] is the value-name of a successful ϕ : “Trust is the act of fulfilling the inherent expectation of values in mutuality” (MVD p. 73). Justice is the operation; trust is the value it deposits (MVD p. 72). The knowledge order’s telos [**data**] is fourfold: “The goal of jeevan is happiness, and the human goal is resolution, prosperity, fearlessness, and coexistence” (JV p. 165) — what the justice cycle *evidences* when it closes, scaled to humankind as undivided society (*akhand samaj*, SB pp. 246–247). These are provisions of coexistence, not automatic outcomes: delusion, accumulation detached from right-use, and fear-based sociality remain live failures (KD ch. 1).

4.10 Laws and propositions

The laws and propositions the texts assert, read through the reconstruction:

- **Universal recognition-fulfilment** — every unit recognises and fulfils at the level of its order (JV p. 69); material, pranic, and animal conduct is definite, while human reflexive judgement may mis-evaluate under *bhram* and becomes definite in awakening.
- **Complementarity, not struggle** — the coupling engine is mutual offering and acceptance, “there is no conflict in existence” (JV p. 157). The texts reject conflict and the pursuit of “more” (JV p. 41) as the engine; the reconstruction accordingly models complementarity rather than competition — a choice of telos, not a proof that optimization is unrepresentable.
- **Assembly by complementary need** — κ ’s trigger: hungry/overfull bonding (MVD p. 8), atoms into molecules into structures (JV p. 67), gravitation itself read as bonding participation (JV p. 150; KD 3.8).
- **Persistence $\Leftrightarrow$ fulfilment** — §4.5.
- **Transmission by order** — §4.5.
- **Closure / iteration** — κ ’s output is a unit, so the schema re-applies; this organizes the tiered hierarchy.
- **Self-evidencing closure** — complete knowledge (*paripoorna gyan*, SB p. 116) requires the reflexive cycle to close in evidenced conduct and transmissible tradition (MVD p. 12).

Four propositions the texts also assert follow. The **tiered hierarchy** (particles $\rightarrow$ atoms $\rightarrow$ molecules $\rightarrow$ planets; cells $\rightarrow$ organisms; individual $\rightarrow$ family $\rightarrow$ undivided society, MVD p. 161) is asserted by the texts and organised by κ ’s iteration. **Stability at the ends, dynamism between** — O is state-complete and existence stable, while units develop definitely within that stability (MVD p. 5, p. 12). The **human knowledge-order discontinuity** is a

conditional consequence of the reflexive model (§2.3): material, pranic, and animal operations execute definitely through constitution, seed, and species-conformance; a developed human body permits the full reflexive circuit, while *bhram* can route that circuit through a body-as-self model and make μ diverge from ε . Human assembly is therefore the unique unfinished tier — “This comprises only humans, who are yet to achieve orderliness” (JV p. 47) — not because its faculties are missing, but because their evaluation and exercise can be misdirected. **Boundedness of extraction** — a unit’s couplings with lower-order units are governed by right-use bounded by regeneration (MVD p. 264; KD 3.13); extraction beyond regeneration is an unfulfilled coupling that destabilises the containing assembly (JV p. 77).

4.11 Time as duration of activity

Time (*kaal*) enters the reconstruction not as a background substance or cause but as a **measure of activity**: “Duration of an activity is referred to as time” (MVD p. 34); “time is measured by the duration of an activity” (SB p. 65). Let $\mathbf{E}$ be the category of unit-executions and let $(\mathbf{D}, \oplus, \emptyset, \leq)$ be an ordered commutative duration monoid. A duration grading $\delta : \mathbf{E} \rightarrow \mathbf{D}$ assigns each execution e a duration and preserves sequential composition:

$$\delta(\text{id}_u) = \emptyset \qquad \delta(e_2 \circ e_1) = \delta(e_1) \oplus \delta(e_2)$$

The identity is the no-change execution of a **unit** u ; it is not *satta*. $\emptyset$ is outside $\mathbf{E}$, since $\emptyset$ is not a unit and performs no activity (§2.1). Consequently $\delta(\emptyset)$ is undefined rather than zero. *Kriya-shunya* entails *kaal-shunya* because duration has no application where there is no activity, not because $\emptyset$ is a zero-duration event. $\emptyset$ ’s presence “at all times” names its omnipresence with every temporally active unit, not temporal becoming in $\emptyset$ (JV p. 42; [Nature of Time](#)).

The equation above grades **sequential** execution only. Coupled units may act concurrently, and their elapsed duration is not obtained by summing every participant’s duration; parallel composition requires interval overlap, a supremum, or another order-appropriate synchronization rule. A clock is a distinguished recurring unit-activity used as a reference. Duration is therefore activity-grounded ontologically but clock-compared numerically. The further claim that studiable causal efficacy belongs to currently executing activity, while past and future are retained as traces, anticipations, and numerical relations, is an interpretive temporal model developed in [Nature of Time](#), not a theorem of the monoid grading.

Time does not select a path, supply energy, or orient activity toward completeness. It records how long an execution or transition lasts. A progression orders states toward A2; *kaal* measures the activity carrying a unit between or within those states. An atom may cycle in natural-state motion, and a deluded *jeevan* may repeat activity, accumulating duration without *vikas* or *jagriti*. The *kaal/rachna/vistar* taxonomy and numerical conventions remain taught data (MVD p. 196).

4.12 The natural state as an attractor: least excitation

The natural/excited distinction (§4.5) admits a **conditional Lyapunov analogy**, but not a universal least-action law. For each order o , let X_o be an admissible state space, $N_o \subseteq X_o$ its natural-state set, and $\Phi_o(t, x)$ the evolution of a unit under specified conducive conditions. For the continuous reading take $t \in \mathbb{R}_{\geq 0}$; a discrete sequence may be used where no differentiable flow is warranted. Let $K_o : X_o \rightarrow \mathbb{R}_{\geq 0}$ measure deviation from N_o :

$$\begin{aligned} K_o(x) &= 0 \quad \text{iff} \quad x \in N_o \\ K_o(\Phi_o(t, x)) &\leq K_o(x) \quad \text{during an unforced relaxation episode} \end{aligned}$$

N_o is an invariant natural-motion regime, not inactivity: activeness continues when $K_o = 0$. The inequality is local and conditional. Fresh perturbation, adverse environment, interaction, decomposition, or human error may increase K_o ; the texts do not supply a global monotonicity theorem. The natural/excited axis is also independent of the compound/mixture axis. A mixture may remain naturally stable without closing a new boundary, and a compound may be excited after it forms.

Natural-state restoration must be separated from development across orders. Relaxation maps a state toward N_o ; it does not by itself move every unit toward constitutional completeness. A guarded partial transition $T1 : N_{\text{material}} \rightarrow X_{\text{sentient}}$, controlled by the still-unknown selection predicate σ_{T1} , represents the exceptional atomic passage to constitutional completeness. Most units continue in their order-relative natural regimes. $T1$ is an attained irreversible threshold in the texts, not an asymptote that every trajectory approaches.

Minimizing K_o is mathematically a form of bounded, constrained optimization. What distinguishes the darshan is not the absence of optimization mathematics but the meaning of the target: restoration to an order-relative qualitative natural set, not indefinite increase of one quantity. At the knowledge order no single scalar is adequate. Relationship fulfilment, justice, faculty harmony, conduct-evidence, and social participation form a vector or satisfaction preorder (§§4.13–4.15); improving one coordinate cannot silently compensate for violating another.

Hamilton's principle supplies a contrast, not the governing model. It selects stationary histories of an action functional under boundary constraints; the action is not generally a conserved quantity, and time symmetry or conservation depends on the particular Lagrangian and its symmetries. Lyapunov relaxation instead describes forward local stability. Neither formalism is stated in the primary texts. The textual claim is natural/excited regulation; K_o , Φ_o , and the attractor reading are mathematical analogies. Nor need development oppose thermodynamic entropy: local organization may increase while total entropy also increases. The open questions are which physical variables, if any, realize K_o , and whether the analogy yields testable dynamics rather than retrospective description.

4.13 Knowing, awakening, and the harmonies of *jeevan*

The knowledge order brings two operations into full human expression before the recognition and fulfilment already present below it. Let v be **knowing** — adequate conception of a unit, relation, or principle — and β be **believing or accepting** that conception as relevant to one's own activity. These are activities of *jeevan*, not faculties created by instruction or practice. Humane conformance then has the typed sequence

$$v \rightarrow \beta \rightarrow \rho \rightarrow \phi \rightarrow \mu \rightarrow \text{mutual satisfaction}$$

The first three orders enact $\rho \rightarrow \varepsilon \rightarrow \phi$ definitely through constitution, seed, or species. A human can know without accepting, accept without recognising a present relationship, or recognise without fulfilling it; these broken composites are formal locations where *bhram* disrupts coherence. Projection (*paravartan*) carries resolve, desire, and thought outward into conduct; reflection (*pratyavartan*) returns result and evaluation through the faculties. Awakening is not a new substance or newly activated inventory but increasing coherence of this recurrent circuit under mediative *atma* (KD 3.12; JV pp. 70, 92).

Let π denote practice. Its type is deliberately not $\pi : \text{MissingFaculty} \rightarrow \text{Faculty}$; it acts on coherence and evidence:

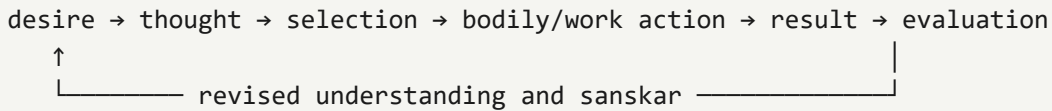
$$\begin{aligned} \pi : \text{CoherenceState}(v, \beta, \rho, \phi, \mu) &\rightarrow \text{CoherenceState}(v, \beta, \rho, \phi, \mu) \\ \text{Activities}(j) &= 10 && - \text{invariant under } \pi \end{aligned}$$

Study can correct conception and belief; repeated practice can expose a mismatch between μ and ε , stabilise right recognition and fulfilment, and make understanding reproducible in conduct. Neither changes the ontological inventory of *jeevan*.

The four *jeevan* values are order-sensitive harmonies, not quantities in one utility scale. Write $H = (h_{mv}, h_{vc}, h_{cb}, h_{ba})$ for accordance between *mun-vritti*, *vritti-chitta*, *chitta-buddhi*, and *buddhi-atma*. The texts name their evidenced states happiness, peace, contentment, and bliss respectively (JV p. 138). The tuple is read by a componentwise or constraint preorder, so a gain in one harmony does not license damage to another. T2, activity completeness, is the closure of knowing and realisation through the faculty circuit; T3, conduct completeness, is the stable projection of that understanding into relationship, work, and tradition. The conditions are textual doctrine organized by the model, not derived from A1 and A2 alone.

4.14 Karma, result, and conduct-evidence

At the knowledge order the universal effort–motion–result triad acquires intentional articulation. A compact karma cycle is



This is not a deterministic pipeline: capacity, ability, receptivity, bodily conditions, other units, and environment mediate the transition from intention to result. Bodily conditions constrain what can be expressed; they do not determine which faculties belong to *jeevan*. Reflexive evaluation μ compares the realised result and value against the recognised relationship and definite relation-evaluation ε ; it may confirm understanding or expose over-, under-, or mis-evaluation. Repeated cycles alter *sanskar*, which conditions later projection without making human conduct constitutionally fixed.

Madhyasth Darshan's evidence condition closes the epistemic and practical sides. A conception is not complete merely because it is internally consistent: understanding must become reproducible in experiment or work, relationship and behaviour, and realisation, and must remain transmissible without contradiction (MVD pp. 12, 336; JV p. 49). Formally, authenticity is a coherence condition across v , β , ρ , ϕ , μ , and τ ; success in one domain cannot substitute for failure in another. This is why conduct-evidence belongs inside the model rather than as an external test appended after it.

4.15 Human goals and the dimensions of universal order

The individual cycle scales into family and social participation through a vector of non-substitutable domains. Let

$S = (\text{education-sanskar, justice-security, production-work, exchange-reserve, health-restraint})$

These five dimensions are the taught institutional content of universal order (JV p. 141). The formal claim is only that social closure requires compatible fulfilment across all five: education carries understanding, justice secures relationships, production provisions prosperity, exchange and reserve coordinate right-use and continuity, and health maintains the body through which *jeevan* evidences conduct. A society cannot compensate for failed justice by producing more goods, or for failed education by stronger enforcement; S is therefore assessed by constraints or a partial order, not one maximand.

The four human goals name outcomes at connected scales: resolution in the person, prosperity in the family, fearlessness in society, and coexistence in universal order (JV pp. 61, 165). They do not arise automatically from assembling institutions. They are evidenced when the knowledge-order cycle (§4.13), karma and authenticity (§4.14), and the five dimensions close together and transmit through humane tradition. This supplies the substantive content that the conduct-compound analogy of §4.4 previously left abstract.

4.16 Causation as conditioned activity and result

The reconstruction does not assign efficient causation to O. O provisions the invariants of A1 while remaining actionless; particular changes belong to units in mutuality. A local causal episode is therefore a typed execution in which a unit's effort and motion, its coupling partners, and its environment jointly condition a result. Since effort, motion, and result are inseparable aspects rather than three detachable events (MVD p. 105), the model does not reduce causation to a single preceding impact:

(unit state, C_o, environment, capacity) —activity—► result state

The result may become a condition for a later execution, giving causal composition without creation from nothing. Duration δ grades the execution but does not cause it; A2 orients admissible development but does not determine every local result. In the material, pranic, and animal orders the transition is constitutionally definite under its conditions. In the human knowledge order intention, bodily expression, other persons, reflexive evaluation, and *sanskar* make the same structure capable of misdirection only when *bhram* enters (§4.14). This separates ontological provision, local efficient activity, temporal measure, bodily constraint, and teleological orientation instead of collapsing them into one notion of cause.

5. The recovery audit

The kernel is adequate only if the reconstructed structure recovers what the texts assert. The audit therefore keeps the two status axes of §1 — **provenance** and **epistemic standing** — separate, with the within-model **role** as a third column. Provenance records where a claim comes from; role records what it does inside the reconstruction; standing records what the formalisation establishes. A theorem below is only a result within the model.

Notion	Provenance	Model role	Epistemic standing
O, U, saturation	textual assertion	primitives	open metaphysical posits
A1	textual synthesis	axiom: activeness, recognition, regulation	adopted premise
A2	textual synthesis	scoped orientation axiom	adopted teleology
C_o, κ	formal reconstruction	typed complementarity and closure	established only within model
Reflexive turn	formal reconstruction	interpretive sentence model	open posit conditioned on T1

Notion	Provenance	Model role	Epistemic standing
Activeness / triad	textual assertion	A1 consequence + taught form	conditional theorem + data
Conservation	textual assertion + reconstruction	model postulate	not derived from A1/A2
No-isolation / coupling	textual assertion	premise for universal coupling	adopted premise
Bearer profile B(u) and aspect map M(u, c)	textual assertion + formal reconstruction	definite carrier with relational manifestation	carrier commitment + taught aspect content
Relationship / association	textual assertion + reconstruction	typed human couplings	definition within model
Value kinds	textual assertion	coupling content	taught data
Recognition / relational evaluation / fulfilment	textual assertion + reconstruction	definite coupling operations ρ , ε , ϕ	defined within model; order content taught
Capacity / ability / receptivity	textual assertion	gates on fulfilment	taught data
Compound / mixture	textual assertion + reconstruction	closure outcomes	typed definition
Human conduct-compound	mathematical analogy + data	social closure model	conditional analogy
Natural / excited	textual assertion + reconstruction	conformance distinction	defined within model
Lyapunov family K _o	mathematical analogy	local relaxation model	conditional and unvalidated
Time δ	textual assertion + formal reconstruction	sequential duration grading	encoding; not a cause
Conditioned causation	textual synthesis + reconstruction	typed activity-to-result transition	formal organization, not a complete physical law
Transmission τ	textual assertion + reconstruction	recurrence of method	form reconstructed; carriers data

Notion	Provenance	Model role	Epistemic standing
Planes and T1–T3	textual assertion	transition types	taught data organized by A2
<i>Jeevan</i> , body expression, and <i>bhram</i>	textual assertion + reconstruction	sentient carrier, asymmetric expression map, human error condition	posit + pairing/interface data
Practice π	textual synthesis + reconstruction	coherence and evidence update, faculty-inventory invariant	formal organisation, not faculty production
$v \rightarrow \beta \rightarrow \rho \rightarrow \phi \rightarrow \mu$	textual assertion + reconstruction	achieved knowledge-order cycle	formal organization of taught stages
Faculty harmony H	textual assertion + reconstruction	non-scalar value state	taught mapping; formal preorder
Karma cycle / authenticity	textual assertion + reconstruction	action-result-evidence loop	formal organization, not derivation
Justice and trust	textual assertion + reconstruction	closure operation and value	definitions + taught content
Human goals and five dimensions	textual assertion + reconstruction	social outcome and domain vector	taught content organized by constraints
Human knowledge-order discontinuity	interpretive reconstruction	conditional consequence of body-supported reflexivity plus <i>bhram</i>	conditional theorem
Tiered hierarchy	textual assertion + reconstruction	organized by typed closure	doctrine, not theorem from κ alone

The **irreducible remainder** stays data and is named plainly: the triad's exactly-three aspects, order-specific manifestations, value kinds, faculty inventory and harmonies, transmission carriers, completeness stages, human goals, five social dimensions, the selection predicates for T1 and *jeevan*–body pairing, and the physical realization of the body-expression interface. The kernel localises rather than eliminates this remainder; the modules show where taught content constrains the formal structure without pretending to derive it.

6. The formal spine

The template invites precise notation. Instead of a system of many categories, one dual-facing spine carries the load, with richer constructions kept as optional annotations.

6.1 Two readings: carrier-behaviour and generation

Downward — carrier-coalgebra. A unit is a carrier x together with a context-indexed behaviour map $x \rightarrow F_C(x)$ unfolding its activity, couplings, and aspect-manifestations. The carrier is *not* quotiented by bisimulation and the theory does *not* pass to the final coalgebra — so the carrier exceeds every relational profile through which it becomes evident. This is the co-structuralist dual of the Yoneda principle, and it is what keeps a unit **substantial** and *jeevan* a real entity rather than a functional role, while letting *O* stay actionless: *A1* grounds *that* the behaviour exists and is lawful, contributing no term to the unfolding.

Upward — generation as an initial algebra. As a **formal reconstruction**, compound units of nature can be **encoded** as the algebra generated by κ from primitive units — an initial-algebra / W-type presentation. κ constructs a new bearer and constitution; it does not manufacture four independent property fields. The resulting $M(u, c)$ is then evaluated from the new bearer in each admitted mutual context. This is the paper’s encoding, not machinery the texts contain; within it, definitions over units (value, transmission, evaluation) become **definitions by recursion** on how a unit was built, and mixture is typed *off* the inductive sort, so “composition is not development” reads as a typing fact.

6.2 The two load-bearing constructs

Two constructs carry most of the encoding’s weight. The **closure operator** κ (§6.1) organizes the hierarchy and iterates by construction. The **society gluing colimit** encodes undivided society: families and communities with shared members glue along overlaps, and the gluing is coherent exactly when local value-structures agree on what they share. This is κ at the social tier, and it is the one place the encoding adds a compatibility **observation** rather than merely restating the darshan: the gluing is coherent only if overlapping assignments actually agree — a descent-style condition whose textual bridge the applied study [How Undivided Society Is Established](#) §4.6 supplies.

The compatibility condition this colimit turns on is not a formal hope — the darshan **asserts** it. *Jeevan* “exists in all humans in the same form, with identical strengths and powers”, and “since the goal of *jeevan* is happiness, all humans aspire for happiness” (JV p. 40); “All human endeavours and experimentations are aimed at achieving satisfaction or happiness” (MVD p. 64); “Human kind is one... Human dharma is one” (MVD p. 13). The agreement is conditioned on awakening — “We humans are united in the right and divided in the wrong” (JV p. 17) — so the diagram is compatible and the colimit glues into an undivided whole exactly in the awakened case, while delusion is where local value-structures disagree and the gluing degenerates (P3). Undivided society is therefore not an incidental output but the **telos the whole structure is driven toward**: the four human goals are the shared value-con-

tent every *jeevan* aspires to, and the society colimit closes coherently precisely when that content is recognised across members. The *content* of those shared values remains taught data; that the aspiration is one, and the agreement real in the right, is what the texts assert — the substantive premise the compatibility condition names.

6.3 Optional annotations

Beyond the spine, several richer categorical models sharpen specific clauses without carrying the theory, and are offered as annotations rather than foundations: reduction to physics as a non-faithful forgetful functor (value-distinct acts collapse); delusion as a retract mistaken for an isomorphism (comfort a genuine part of fulfilment, not the whole, MVD Ch. 4); right-use as a natural transformation uniform across domains; graded fulfilment as enrichment over a satisfaction preorder, fibred over capacity; the activity triad as a traced joint form and κ -bookkeeping as a Petri net; the faculty cycle as a projection–reflection sketch (KD 3.12); and the actualisation of sentience at T1 as a guarded partial morphism — the *Saturation-Reflector* reading whose selection predicate is taught data, developed in [The Ontology of Coexistence](#) §§1.2, 1.6.1 (open problems in §6.2.1). Time is a grading of **unit-executions** over $(D, \oplus, \emptyset)$, with $\delta(\text{id}_u) = \emptyset$ and no value assigned to O (§4.11). Natural/excited regulation is an order-indexed Lyapunov family under specified conditions, kept distinct from the guarded T1 transition (§4.12). Each construction is conditional, and none is a theorem about reality.

7. The structure at every tier

Element	Material	Biological	Animal	Knowledge
Unit	particle, atom, molecule	cell, plant	animal (body + <i>jeevan</i>)	human (body + <i>jeevan</i>)
Four aspects	real bearer; reflected form and definite effects in mutuality	same, through living composition	same, through species-limited bodily expression	same; reflexive judgement can misread the definite mutuality
Saturation (A1)	inherent energy and regulation in unit through co-location	same	same	same

Element	Material	Biological	Animal	Knowledge
Regulation	law → structural conformance	law → seed conformance	law → species conformance	law → <i>sanskar</i> conformance; inward regulation; justice
Relationship	bonding mutuality	nutrient/season	herd, lineage	the definite human relations, with purposes (JV p. 109)
Value	utility + art (constant)	utility in food-chains	utility + sensitivity	<i>jeevan</i> , human, established, expression; μ evaluates
Recognition-evaluation-fulfilment	$\rho \rightarrow \epsilon \rightarrow \phi$, definite	$\rho \rightarrow \epsilon \rightarrow \phi$, definite	$\rho \rightarrow \epsilon \rightarrow \phi$, definite	μ judges ϵ ; justice when the cycle closes, mis-evaluation under <i>bhram</i>
Time	duration of activity	duration of activity	duration of body and <i>jeevan</i> activity	duration plus numerical reckoning and planning
Activity triad	definite; excitation still complementary	definite	definite; expression body-limited	all ten activities inherent; routes through understanding; <i>bhram</i> possible
Assembly schema	literal compound-closure	compositional organization	body via lineage; herd association	formal conduct-compound analogy
Persistence	natural state	climatic balance	species conditions	fulfilment; mutual satisfaction
Transmission	constitution	seed (<i>rachna vidhi</i>)	lineage/species	education- <i>sanskar</i>

Element	Material	Biological	Animal	Knowledge
Knowing and evidence	not reflexive	not reflexive	bodily expression of sensation, selection, and hope	$v \rightarrow \beta \rightarrow \rho \rightarrow \phi \rightarrow \mu$; practice evidences coherence in work, behaviour, realisation
Constraint / failure	excitation without mis-evaluation	death/decomposition without mis-evaluation	bodily or brain constraint; species stress	body-identification $\rightarrow$ mis-evaluation $\rightarrow$ exploitation, war, ecological overdraw

The column-shape repeats four times — the precise sense in which the structure is universal, asserted for each order rather than carried by metaphor from human life. Each row is richer in the knowledge order but never absent below it: value is “the essentiality of units in every plane and order” (SB p. 50), and relational evaluation is definite throughout the first three orders. What first appears with humans is reflexive judgement of value — and, under *bhram*, the possibility of error.

8. What the structure is not

The most natural modern glosses quietly replace the structure with something else. It is **neither a static property bundle nor pure relationism**: $B(u)$ is a real definite bearer, while $M(u, c)$ records how its inseparable aspects are reflected and evidenced in mutuality. It is **not driven by indefinite increase**: the darshan does not make unbounded “more” the telos — conduct in the first three orders is definite (JV p. 113), object-usefulness is constant (JV p. 123), and “more” is the diagnosed pathology (JV p. 41). A terminal state can, of course, be *represented* by minimising distance to it; the claim is not that optimisation is unrepresentable but that **indefinite quantitative increase is not the goal**, so this reconstruction encodes completeness as a qualitative terminal condition rather than an unbounded scalar objective. It is **not struggle or selection**: the assembly engine is complementarity, explicitly opposed to conflict and survival-of-the-fittest (JV p. 157); decomposition is the *loss* of structure when relations go unfulfilled, not its designer. It is **not bottom-up emergence alone**: capacities are provisioned in existence and manifest when composition reaches the right tier (JV p. 59), and O energises every tier at once — a purely emergentist reading drops O and turns energisation into magic. And it is **not value-as-epiphenomenon**: value is the content of fulfilment at every tier (SB p. 50), and the human innova-

tion is a fallible second-order judgement on definite mutual value, located in *jeevan*, “not bodily or mechanical” (JV p. 39).

9. Relation to other frameworks, and uses

Stated this generally, the structure invites comparison — useful chiefly for locating what is distinctive.

Framework	Shared structure	Where it differs
Holarchy (Koestler)	every unit is whole and part	no medium O, no value in relations, no transmission
Autopoiesis (Maturana & Varela)	persistence by continuous self-production	living systems only; the structure runs from atoms to societies and grounds persistence in fulfilment
Multilevel selection	nested units persisting or failing as wholes	selectionist engines are competitive and stochastic; here the engine is complementarity and definite
Thermodynamic self-organisation	larger structures from flows between non-equilibrium units	dissipative structures need free-energy throughput; here units are energised constitutively by saturation
Category-theoretic composition	relationships as morphisms; assemblies as colimits; tiers as a partial order	notation for κ and iteration, but silent on which compositions occur; complementarity is the selection rule the notation lacks

Two commitments remain genuinely distinctive: the medium O as a constitutive, energising, regulating presence at every tier at once — a precondition *of* physics, not an aether within it; and value as the universal content of relations, with fallible reflexive judgement appearing only in deluded human living. Frameworks that drop the first must find energy and regulation inside the units; frameworks that drop the second must reconstruct value as preference or fitness.

The uses follow the forward and reverse readings of §1. For **human organisations**, a deliberate assembly persists only if its relationships are recognised and their values actually fulfilled — not merely incentivised — and it transmits its understanding to incoming members; an organisation that fulfils values but does not transmit decays on turnover, one that transmits rules without fulfilment holds only as long as enforcement does ([How To Form Self-Sustaining Organizations](#); [How Undivided Society Is Established](#)). For **ecology and economics**, boundedness of extraction turns sustainability from an ethical add-on into a stability condition. For **AI and value alignment**, graded fulfilment is an argument for multi-level objectives over one scalar reward — optimising a single number maximises the wrong

thing, which is the darshan's own diagnosis of "more". And for the **study of the darshan itself**, the kernel is a checklist for reading any passage: identify the unit, the tier, the couplings, the expectation profile, the value, the conformance regime, and the transmission carrier, and much apparent repetition resolves into the same move at different tiers.

10. Limitations and open problems

Formalisation is interpretive. The symbols (O, U, A1, A2, B, M, C_o, κ , ν , β , ρ , ε , φ , μ , π , express, Bhram, τ , δ , X_o, N_o, Φ_o , K_o, σ_{T1} , H, S) do not appear in the texts; they are this paper's compression, warranted by the citations. Every reconstruction is this paper's construction.

The heavy posits, named. The simplification concentrates the theory's contestable weight onto A2's teleology, the mediative regulation asserted in A1, and two selection predicates the kernel hosts but does not derive: which atomic configuration reaches constitutional completeness (T1), and which *jeevan* animates which body (the expression pairing, §4.8). A physicist critic will press exactly there. The reflexive turn, duration grading, Lyapunov family, aspect-manifestation map, and body-expression interface organize textual claims but do not establish their physical realization.

The residual premise. Universal coupling rests on the **no-isolation premise** (JV p. 43; §2.4) — a textual assertion the reconstruction adopts rather than derives; that units couple at all does not follow from A1 alone. This is the one place the kernel leans on a premise beyond its axioms.

Substantiality and relational description. Category-theoretic and other structural descriptions are relational by default, so an incautious formalisation tends to re-describe a unit, especially *jeevan*, as a functional role. The carrier-coalgebra reading (§6.1), with carrier-primacy stipulated (§2.1) and the final coalgebra refused, holds the anti-reductionist line by commitment. $M(u, c)$ therefore indexes evidence by mutuality without claiming that the bearer is nothing beyond all possible relations. The exact metaphysical relation between bearer and complete relational profile remains a boundary of the tools.

Translation risk. Quotations from MVD, SB, and JV follow the published English translations; several key terms (*vyavastha* as "orderliness", *samprikt* as "saturated") carry translators' choices. Citations to *Manav Karm Darshan* (KD) are to a machine-assisted **working** translation, not a published one; they are given as corroboration only where MVD/SB/JV already carry the claim, and KD-only material should be verified against the Hindi page images before release. The corpus beyond these four — *Vyavaharatmak Janvad* and *Anubhavatmak Adhyatmvad* — is not yet cited.

No empirical contact. Nothing here yields a measurement or prediction of *jeevan*, coexistence, or constitutional completeness. Elegant diagrams can create an illusion of proof; the conditional framing and this list are the safeguard.

Open problems. Formal countermodels establishing the independence of A1 and A2; discharging or confirming no-isolation; identifying the order-relative relations C_o ; operationalising $M(u, c)$ without collapsing bearer into observation; a mechanised recovery audit; deriving over-, under-, and mis-evaluation from the $\mu \neq \varepsilon$ condition; modelling the animal and human body-expression affordances, including how neural development or injury constrains evidence without altering *jeevan*'s inventory; specifying how practice changes coherence without creating faculties; testing the society-gluing compatibility condition; specifying concurrency in the duration model; finding physical variables and domains for any K_o ; relating local organization to thermodynamics without positing an anti-entropic law; formalizing the non-scalar preorders H and S ; and identifying the selection predicates for T1 and *jeevan*-body pairing. Comparison with stationary action and gradient flows remains open comparative work.

11. Conclusion

Read from first principles, Madhyasth Darshan's ground is coexistence itself: one actionless medium and countless real units, bound by saturation. A unit is a definite bearer, not a four-field property bundle; form, *guna*, *svabhav*, and *dharma* become definitely evident through mutuality without being created by observation. A1 provisions activeness, recognition, and inherent regulation; A2 supplies order-relative orientation toward completeness; typed complementarity and closure organize material compounds, biological composition, and the declared analogy of humane social order. Time grades the duration of unit-activity without causing it. Natural/excited restoration admits a conditional stability model, kept separate from the selected transitions of development. At constitutional completeness the interpretive reflexive model opens *jeevan*: all five faculties and ten activities belong to it. Animal bodily organization limits expression; in humans, *bhram* and body-identification can make reflexive judgement diverge from definite mutuality. Study supplies understanding, while practice tests, evidences, and stabilises coherence rather than creating faculties. Karma makes that understanding answerable to result and conduct; faculty harmonies give non-scalar content to happiness, peace, contentment, and bliss; the five social dimensions carry the same requirement into universal order. The reconstruction is strongest when every claim keeps its textual provenance, within-model role, and epistemic standing visible — especially where a mathematical analogy organizes an open metaphysical posit without establishing it.

References

Madhyasth Darshan (primary sources)

- **MVD** — Nagraj, A. [*Madhyasth Darshan \(Manav Vyavahar Darshan\)*](#). English version 4.0. Cited: state-complete/state-dynamic and the evidence chain (p. 12); *gyan* as intelligibility-ground (p. 11); mediative Omnipotence and nucleus (p. 26); basic

impulsion (p. 40); hungry/overfull atoms (p. 8); composite atoms, mixture and compound (p. 42); relationship vs association, capacity–ability–receptivity (pp. 61–62); joint form of the triad (p. 105); mutuality pressure in results (p. 114); *atma* and orbital faculties (p. 78); ascending/frustration (p. 79); frustration as yearning for restfulness (p. 104); justice as closure (pp. 311, 336); trust as fulfilment (pp. 72–73); evaluation failures (p. 38); insentient/sentient activity bounds (p. 33); inward regulation (pp. 77, 277); ten-tier orderliness (p. 161); regeneration bound (pp. 263–264); humankind and human *dharma* as one (p. 13); all human endeavour aimed at satisfaction and happiness (p. 64); time as duration of activity (p. 34) and the *kaal/rachna/vistar* taxonomy (p. 196). Page numbers refer to this edition.

- **SB** — Nagraj, A. [*Samadhanatmak Bhautikvad \(Resolution Centred Materialism\)*](#). English translation by Rakesh Gupta. Cited: unit as whole with environment, orderliness with *ness*, natural and excited state (pp. 13–14); motion-path as regulation line and boundary (p. 58); unit + energy fullness = activeness (p. 69); completeness goals of the triad (pp. 58, 71); regulation as law with recognition provision (p. 57); essentiality as value (p. 50); constitutional completeness and inexhaustibility (pp. 55, 59, 61); composition is not development (pp. 75–76); order conformance regimes (p. 236); planes and T1 irreversibility (pp. 51–52, 92); undivided society and human *dharma* (pp. 246–247); complete knowledge (p. 116); every unit active and developing for completeness (p. 51); time measured by the duration of activity (p. 65). Page numbers refer to this edition.
- **JV** — Nagraj, A. [*Jeevan Vidya: An Introduction*](#). English translation, version 3.0. Cited: space uniform and omnipresent (p. 31); no provision to separate units from Omnipresence (p. 18); nothing isolated (p. 43); countable entities (p. 45); order conformance (p. 48); understanding flows in tradition (p. 49); *rachna vidhi* in the *pran sutra* (p. 82); no conflict in existence (p. 157); knowing and believing (p. 70); existence neither increases nor decreases (p. 127); universal recognition-fulfilment (p. 69); assembly inclination (p. 67); gravitation as bonding participation (p. 150); six-fold values and established values, including the four faculty harmonies (pp. 43, 108, 138–139); *jeevan* evaluates without external instrument (pp. 39, 139); ten activities across nucleus and orbits (p. 92); body provisioned to evidence understanding (p. 59); alignment of *jeevan* values and human goals (p. 61); five dimensions of universal order (p. 141); human goals (p. 165); definite conduct of the peepal tree (p. 113); commerce and “more” (p. 41); Earth-relationship overlooked (p. 77); humans united in the right, divided in the wrong (p. 17); *jeevan* identical in all humans, all aspiring for happiness (p. 40). Page numbers refer to this edition.
- **KD** — Nagraj, A. *Manav Karm Darshan* (Madhyasth Darshan — Part 2). [working English translation](#) of chapter 3; Hindi source [KD-karm darshan v5.pdf](#). **Machine-assisted working translation — not a published translation; cited as corroboration only, and to be verified against the Hindi page images before release.** Cited: the title-page principle Effort–Motion–Result; the saturation → energy-

endowment → force-endowment → recognition → orderliness chain (3.1); the sense of the triad's three aspects (3.11); the two developments, compositional and atomic (3.2–3.3); hungry/undigested/satisfied atoms and the three completeness stages (3.3); the five state/motion faculties (3.6); force, weight, and gravitation as unit-mutuality (3.7–3.8); projection–reflection (3.12); extraction and regeneration (3.13); the human as body and *jeevan* (3.16); time as duration of activity, the ever-present, and the clock-time critique (3.15); excitation relaxing to natural-state motion (3.5); ninefold karma and result (ch. 1).

Related studies in this collection

- [*The Ontology of Coexistence*](#) — ontological exposition of saturation, definite bearers and relationally evidenced aspects, the regulation ladder, six-fold value taxonomy, orders, planes, and progressions; definite *jada* mutuality; animal bodily limits and human body-delusion; *jeevan* faculty structure and inward regulation (§§1.7–1.9); four human goals and provisions vs achievements (§1.15); Saturation-Reflector reading and T1 latency (§§1.2, 1.6.1; open problems in §6.2.1)
- [*How Undivided Society Is Established*](#) — assembly composition from family through humankind, and the compatibility bridge for the society-gluing colimit (§4.6)
- [*The Epistemology of Coexistence*](#) — evidence chain, *gyan udghatan*, *pramana*
- [*Nature of Time*](#) — *kaal* as duration of unit-activity; the full treatment §4.11 builds on (present-primacy, timeless *satta*, the physics comparison)
- [*How To Form Self-Sustaining Organizations*](#) — the forward use applied to deliberate assemblies
- [*Why Humans Are Not Just Material*](#) — human-tier anthropology
- [*Human Behavior and Society*](#) — conduct and social order

The planned *Restfulness and Least Action*, *Death, Continuity, and Rebirth*, and *Governance, Justice, and Undivided Society* studies are referenced above where their material is deferred.